Shaft designs

Knowns - Gear ratio = 24:9 = GR

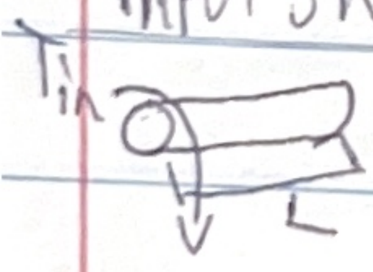
Input Shaft (Power-transmitting)
- Diameter Shaft = 0.125 in
- Length = L = 1 in
- Allowable τ_s for Power-transmitting Shafts = 4000 Psi

Output shaft (Short shaft)
- Diameter shaft = 0.25 in
- L = 1 in
- Torque (T) = 1.25 in-lb
- Speed (N_o) = 1100 rpm
- Allowable τ_s for short shafts = 8500 Psi

Unknowns

Torque in (T_i), N_i (speed in) P_{in} (Power in), P_{out} (Power out)
$\tau_{s_{in}}$ (Torsional stress in), τ_{so} (Torsional stress out)

Input shaft



$\tau_{imax} \leq (\tau_{allow})\left(\frac{\pi}{16}(0.125)^3\right)$

$0.04875\ in\text{-}lb \leq 1.53398\ in\text{-}lb$

D_i is sufficient

$N_{in} = GR(N_o) = \frac{24}{9}(1100) = 2933.33\ rpm$

$T_{in} = T_o/GR = 1.25/\frac{24}{9} = 0.46875\ lb\text{-}in$

$P_{in} = \frac{T_{in} N_{in}}{63{,}000} = 0.0218\ HP$

$D = \sqrt[3]{\frac{80(0.0218)}{(2933.33)}} = 0.0841\ in$

$\tau_{s_i} = T_{in}/Z_p = \frac{(0.46875)}{\frac{\pi}{16}(0.125)^3} = 1222.31\ psi$

Output Shaft



$\emptyset = 0.25$

$$P_0 = \frac{T_0 N_0}{63000} = \frac{(1.25)(1100)}{63000} = 0.0218 (HP)$$

$$D_0 = \sqrt[3]{\frac{80 P_0}{N_0}} = \sqrt[3]{\frac{80(0.0218)}{1100}} = 0.11661 in$$

$$\tau_{so} = \frac{T_0}{Z_p} = \frac{(1.25)}{\frac{\pi}{16}(0.25)^3} = 407.436 psi$$

$$T_{o_{max}} \leq (\tau_{sallowable})(Z_p)$$

$$1.25_{lb\text{-}in} \leq (8500)\left(\frac{\pi}{16}(0.25)^3\right)$$

$1.25\ lb\text{-}in \leq 26.0777\ in\text{-}lb$, D_0 is sufficient

Journal Bearings

Knowns -> $L = 1in$, $\alpha = 20°$

Input shaft -> $T_i = 0.46875\ lb\text{-}in$, $D_i = 0.125$, $N_i = 2933.33\ rpm$

$\tau_{si} = 1222.31\ PSI$

Output shaft -> $T_o = 1.25\ lb\text{-}in$, $D_o = 0.25$, $N_o = 1100\ rpm$

$\tau_{so} = 407.436\ PSI$

Unknowns

P_{V_i}, V_i, P_{V_o}, V_o, F_{t_o}, F_{r_o}

$F_{t_i} = \frac{2T}{d}$, $F_r = F_t \tan(\alpha)$, $V = \omega \pi D$, $P = F/A$, $A = L \cdot D$

Input	Output
Shaft $\varnothing = .125\ in$, Bearing $\varnothing .375$	Shaft $\varnothing .25$, Bearing $\varnothing .375$

$F_{T_i} = \frac{2(0.46875)}{0.375} = 2.5\ lb$ | $F_r = 2.5 \tan(\alpha) = 0.9099\ lb$

$V_i = \pi (0.125)(2933.33\ rev/min) = 95.99\ ft/min$

$\frac{}{12}$

$P_i = \frac{0.9099}{0.375 \cdot (0.125)} = 19.4112$

$V_o = \frac{\pi (0.25)(1100)}{12} = 71.99\ ft/min$

$F_{T_o} = \frac{2(1.25)}{1} = 2.5\ lb$

$P_o = \frac{F_r}{0.25(0.375)} = 9.7056\ PSI$

$F_r = 0.9099\ lb$

$P_{max} = 19.4112$ psi

$V_{max} = 95.99$ ft/min

$PV_{max} = 1863.281$ psi · ft/min

looking at ~~Table~~ Table 6: Application limits

Choose Graphite (dry): 600 psi, 250 ft/min, 1500 psi · ft/min

Bearing thickness



Knowns

$P_{max} = 600$ psi $L = 0.375$

$F_r = F_N = 0.9099$ lb

$A = 2t \cdot L$

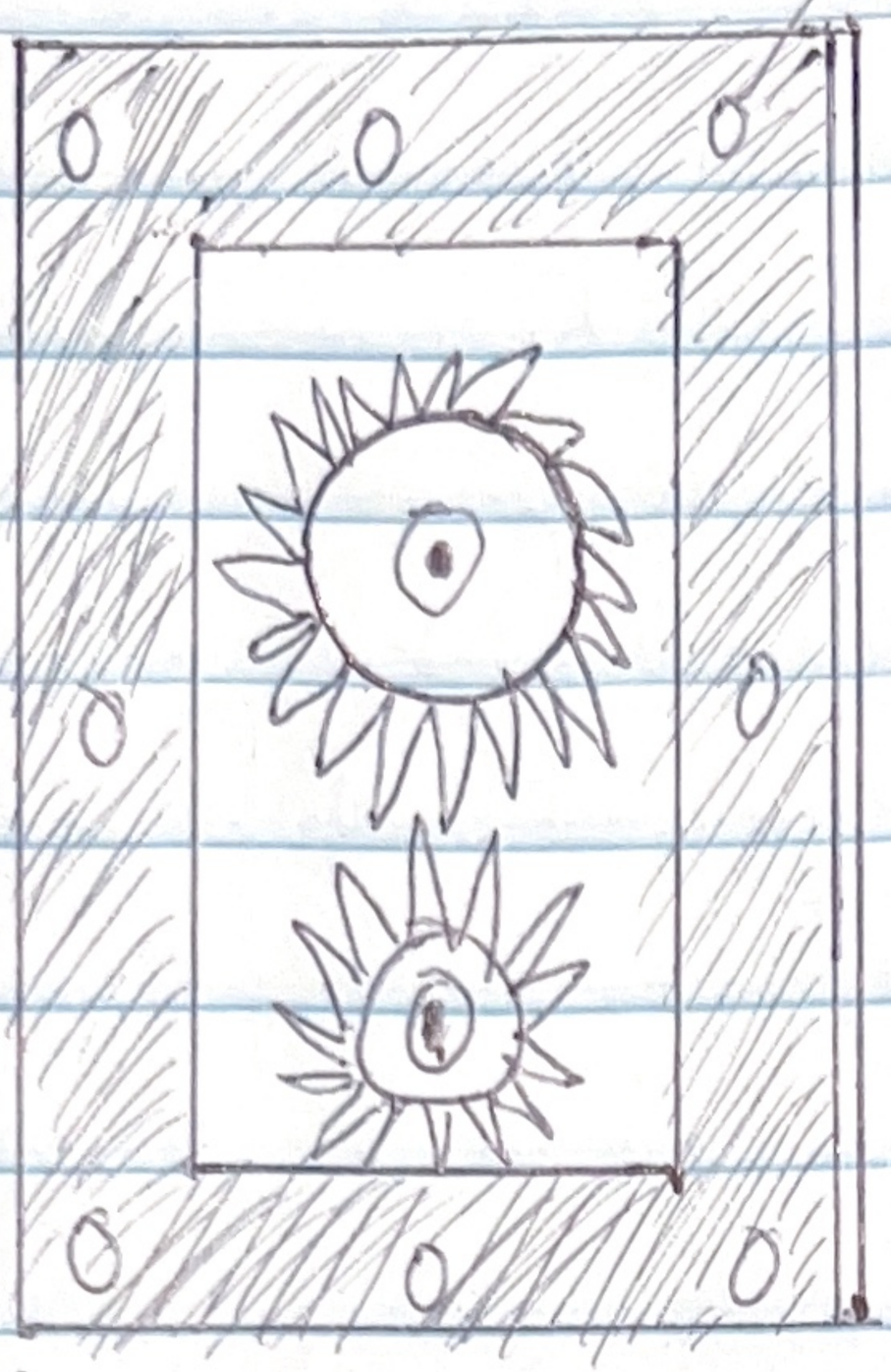
Unknown -> t

$P = \frac{F_r}{A} \rightarrow A = F_r / P$

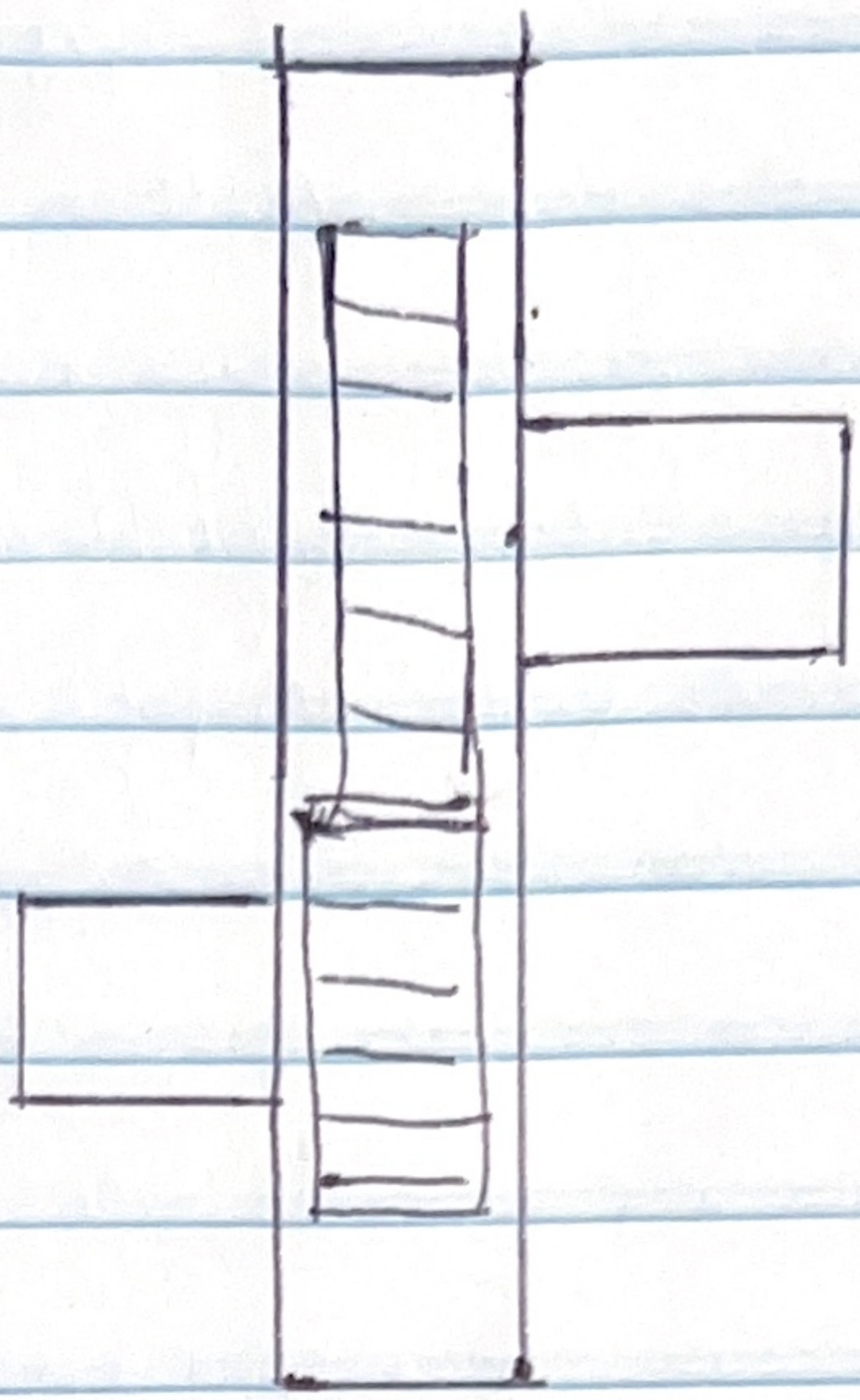
$t = \frac{F_n}{P \cdot 2L} = \frac{0.9099 \text{ lb}}{2(600)(0.375)} = 0.0020$ in

Sketch

screws



Front View



Side View